EE4-55
EE9-AO11

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2021

MSc and EEE PART IV: MEng and ACGI

MEMS AND NANOTECHNOLOGY

Wednesday, 28 April 10:00 am

Time allowed: 3:00 hours

There are FIVE questions on this paper.

Answer Question 1.
Answer Question 2 OR Question 3.
Answer Question 4 OR Question 5.

Question 1 carries 40% of the marks. Remaining questions carry 30% each.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : A.S. Holmes, Z. Durrani, A.S. Holmes

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Information for Candidates

Permittivity of free space:	$\varepsilon_0 = 8.854 \times 10^{-12}$ F/m.
Boltzmann constant:	$k_B = 1.38 \times 10^{-23}$ J/K
Electron charge:	$e = 1.6 \times 10^{-19}$ C.

The Q-factor of an in-plane micromechanical resonator damped by viscous losses in the gas between the moving element and the substrate is given by:

$$Q = \frac{m\omega_0 d}{\mu A}$$

The deflection profile $v(x)$ of a flexure of length l when subject to a transverse load P is:

$$v(x) = \frac{Pl^3}{12EI}(3u^2 - 2u^3)$$

where $u = x/l$ and the other parameters have their usual meanings.

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This question is compulsory

1. a) Consider a spherical Si nanocrystal, with diameter d = 2 nm. Calculate the maximum photon emission wavelength λ in the nanocrystal due to electron-hole recombination. You may use an effective mass for electrons and holes in Si of m_e = 1 and m_h = 0.57 respectively. The rest mass of an electron $m_0 = 0.91\times10^{-30}$ kg, Planck's constant $h = 6.625\times10^{-34}$ J.s, the speed of light $c = 3\times10^{8}$ m/s, and the band gap in bulk Si, E_g = 1.1 eV. [5]

b) Figure 1.1 shows schematically the cross-section through a Si nanowire FET using a planar gate. The Si nanowire length L = 100 μm and Si core radius R = 10 nm. An oxide shell with thickness t_1 = 5 nm surrounds the nanowire. The back-gate is covered by a planar oxide layer of thickness t_2 = 10 nm. The gate capacitance for the device can be approximated by:

$$C_g = \frac{2\pi\varepsilon_r\varepsilon_0 L}{ln\frac{2H}{R}}$$

where H is the minimum separation between the nanowire Si core and the back gate.

Calculate the carrier concentration n, per cubic metre, in the nanowire at threshold voltage V_{th} = 0.4 V.

You may use $\varepsilon_0 = 8.854\times10^{-12}$ F/m and $e = 1.6\times10^{-19}$ C. The dielectric constants for Si and SiO_2 are ε_{si} = 11.9 and ε_{ox} = 3.9 respectively.

[5]

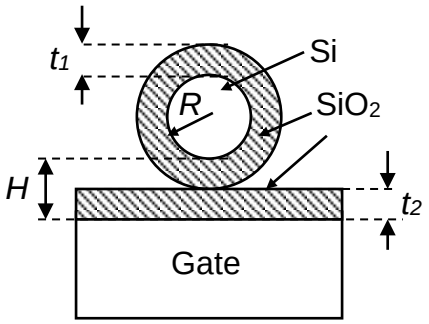


Figure 1.1

c) X-ray irradiation of a crystal at a wavelength λ = 0.3 nm for different angles of incidence is seen to lead to a series of diffraction peaks. Diffraction peaks are seen to arise at angles of incidence to the normal of 64.3° and 30°. Calculate the corresponding atomic plane separation. [5]

Question 1 continues on the next page.

Question 1 continued.

d) Using suitable diagrams, explain briefly the operation of a double barrier heterostructure resonant tunnelling diode. [5]

e) A square silicon nitride membrane is to be fabricated by anisotropic etching a (100) silicon wafer which has been coated both sides with silicon nitride. An optical monitor is used to determine the end-point of the etching process. If the membrane is to have a side length of 1.0 mm, the silicon wafer is 525 μm thick, and the etchant has an anisotropy of 35, what size of opening should be made in the nitride layer on the back side? Also, if the tolerances on the wafer thickness and anisotropy are ±25 μm and ±5 respectively, what range of membrane sizes can be expected? [5]

f) Explain briefly, with the aid of diagrams, the operational principles and typical geometries for material bimorph and shape bimorph electrothermal actuators. Highlight the differences in functionality between the two types and any advantages one type may have over the other. [5]

g) This question relates to the scaling behaviour of an in-plane micromechanical resonator under two scenarios:
   (i) all linear dimensions are scaled in the same way;
   (ii) the mechanical layer thickness and the gap between mass and substrate are held constant as the lateral dimensions are scaled.

   Assuming the Q of the resonator is dominated by the viscous losses in the air between the mass and the substrate, derive scaling laws for the resonator Q under the two scenarios described above. [5]

h) Starting from the deflection profile provided in the *Information for Candidates*, show that a flexure with fixed span l will, when deflected laterally by an amount v_l, will be subject to an axial strain given approximately by:

$$\varepsilon \approx \frac{3}{5}\left(\frac{v_l}{l}\right)^2$$

[5]

End of Question 1.

Answer EITHER Question 2 OR Question 3

2. a) The electron wave function $\psi_{100} = A \exp(-Zr/a_0)$ represents the $|1s\rangle$ state of a hydrogenic atom in spherical coordinates (r, θ, φ). Here, Z is the atomic number and a_0 is the Bohr radius. Derive an expression for the amplitude A.

[10]

b) Figure 2.1 shows a Si atom, labelled '1', within the Si unit cell. Bonds b_1, b_2, b_3 and b_4 connect this atom to its nearest neighbour atoms.

(i) What is the Si-Si bond length in nanometres?

[4]

(ii) Write down, in the form [*u v w*], fundamental direction vectors for the bonds, assuming these point from atom 1 towards the nearest neighbour atoms.

[4]

c) Hence, construct $|sp^3\rangle$ wavefunctions for the bonds b_1, b_2, b_3, b_4. Your answer should include a detailed derivation for at least one of the bonds. You may use the $|2s\rangle$ and $|2p\rangle$ states of the hydrogenic atom, in spherical coordinates (r, θ, φ), given below:

$$|2s\rangle = \psi_{200} \sim \left(1 - \frac{Zr}{2a_0}\right) e^{\frac{-Zr}{2a_0}}$$

$$|2p_0\rangle = \psi_{210} \sim e^{\frac{-Zr}{2a_0}} r\cos\theta$$

$$|2p_{\pm 1}\rangle = \psi_{21,\pm 1} \sim e^{\frac{-Zr}{2a_0}} e^{\pm i\phi} r\sin\theta$$

[12]

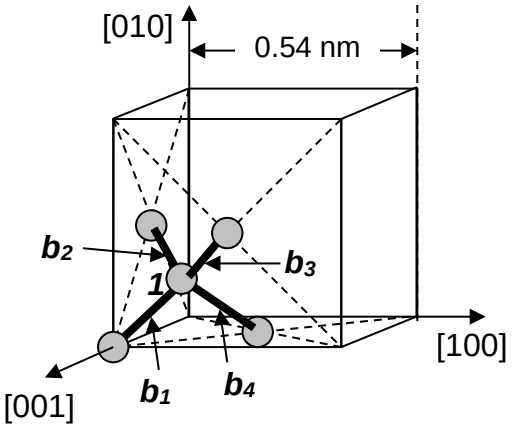


Figure 2.1

3. One dimensional electron waves with amplitudes *A*, *B*, α, β, *C*, and *D*, tunnel across a potential barrier of height V_0 as shown in Figure 3.1 below:

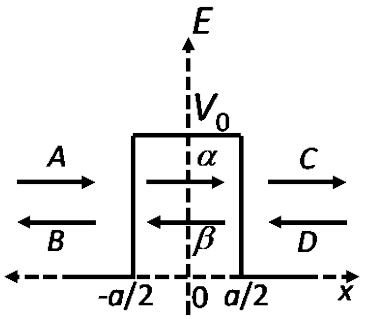


Figure 3.1

The potential energy *E* is equal to V_0 for $-a/2 < x < a/2$ and is zero elsewhere.

a) Write down general expressions for the electron wave vectors k_1 and k_2 in the regions outside, and within the potential barrier, respectively.

[4]

b) Transmission matrices of the electron waves at the boundaries $-a/2$ and $a/2$ are given by:

$$\Gamma_1 = \frac{1}{2k_2}\begin{pmatrix} k\exp\left(k^*\frac{a}{2}\right) & k^*\exp\left(k\frac{a}{2}\right) \\ k^*\exp\left(-k\frac{a}{2}\right) & k\exp\left(-k^*\frac{a}{2}\right) \end{pmatrix}, \quad \text{at } x = -a/2 \qquad \text{Eq. 3.1}$$

$$\Gamma_2 = \frac{i}{2k_1}\begin{pmatrix} -k\exp\left(k^*\frac{a}{2}\right) & k^*\exp\left(-k\frac{a}{2}\right) \\ k^*\exp\left(k\frac{a}{2}\right) & -k\exp\left(-k^*\frac{a}{2}\right) \end{pmatrix}, \quad \text{at } x = a/2 \qquad \text{Eq. 3.2}$$

where $k = k_2 + ik_1$ and k^* is the complex conjugate of *k*.

Using these matrices, write down appropriate expressions relating the wave amplitudes *A* and *B* to α and β. Also write down expressions relating *C* and *D* to α and β.

[4]

c) Hence, derive the transmission matrix for the entire potential barrier.

[10]

d) Calculate the determinant for this matrix. You may simplify using the identity $(c)^4 + (c^*)^4 - 2|c|^4 = -16a^2b^2$, for a complex number $c = a + ib$.

[12]

Answer EITHER Question 4 OR Question 5

4. Figure 4.1 shows an open-loop z-axis MEMS accelerometer fabricated by surface micromachining. Acceleration normal to the wafer surface is sensed through variation in the capacitance between the proof mass and the underlying substrate. The square proof mass, of side length 1 mm, is supported by four serpentine springs, each containing four beams of length 400 μm and width 20 μm. The mechanical layer is 5 μm thick and the gap between the proof mass and the substrate is 2 μm under zero acceleration.

   a) Suggest a fabrication sequence for the device, starting from a silicon wafer. You should name the processes involved at each step but you do not need to describe them in detail. Your answer should include sketches showing cross-sections along *XX'* at each stage of the process. [6]

   b) Assuming each serpentine spring behaves as four flexures connected in series, calculate the stiffness of the suspension. Assume a value of 170 GPa for the Young's modulus of silicon. [6]

   c) Calculate the sensitivity of the accelerometer (capacitance change per unit acceleration) at low acceleration levels. When calculating the capacitance you should ignore the suspension beams and the proof mass perforations. The density of silicon is 2330 kg/m^3. [6]

   d) The readout circuit applies a continuous sinusoidal voltage of 1.0 V rms amplitude to the accelerometer. If the device is tested in a centrifuge, and the acceleration in the +z direction is gradually increased, at what acceleration level will the proof mass snap down to the substrate? (Hint: the applied acceleration will shift the force-vs-gap curve downwards, and you need to calculate the acceleration value for which the stationary point occurs at zero force.) [8]

   e) Measurements made on a prototype device indicate that the suspension stiffness is lower than the value you calculated in part b). If dimensional errors can be ruled out, and the Young's modulus value is correct, what do you think is responsible for this discrepancy? [4]

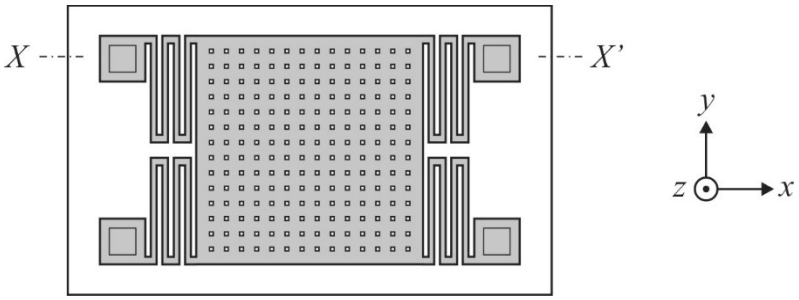


Figure 4.1

5. Figure 5.1 shows an electrostatic actuator which is designed to rotate a MEMS micro-mirror about an axis normal to the wafer surface and passing through (or very close to) the centre of the mirror surface. The device is fabricated on a BSOI wafer with a mechanical layer of thickness $h = 50$ μm. The six suspension beams have length $l = 500$ μm and width $w = 5$ μm, and the mirror has length $m = 50$ μm. Each comb drive has $N = 40$ electrode gaps, each of width $g = 5$ μm.

   a) Explain how the actuator works. Your answer should include a sketch of the device in a deflected state, clearly showing the deflection profiles of the suspension beams and the displacements of the comb drives. Also indicate which comb drives *must* be activated in order to achieve the deflection you show. [6]

   b) Neglecting the fact that the mirror block is wider than the beams that support it, estimate the suspension stiffness experienced by the comb drives on either side when the comb drive displacements on the left and right sides are equal and opposite. You should assume a value of 170 GPa for the Young's modulus of silicon. [8]

   c) Show that, when voltages V_A and V_B are applied to the electrodes labelled A and B respectively, the forces applied to the two sides of the suspension are given by:

$$\pm \frac{N \varepsilon_0 h}{g} V_0 V_D$$

   where V_0 and V_D are the common-mode and differential drive voltages. [6]

   d) By solving the bending equation for a cantilever subject to a transverse end load, show that the end rotation θ is given by:

$$\theta = \frac{3v}{2L}$$

   where v is the end deflection and L is the cantilever length. [6]

   e) Using the results in parts b) to d), calculate the rotation angle of the mirror when $V_0 = V_D = 100$ V. [4]

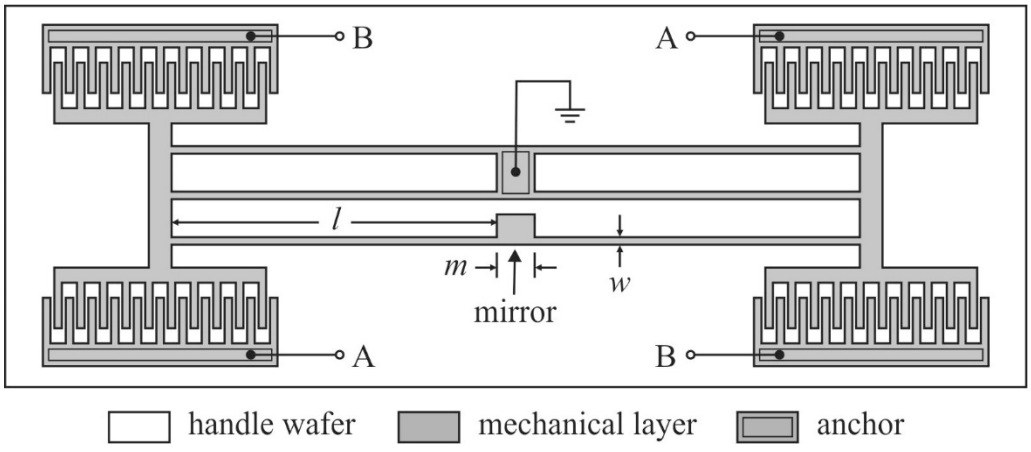


Figure 5.1